

# Module-wise Question Bank (Up to Module 3)

## Module 1: Introduction – Number Theory and Basic Cryptography

- Define Active and Passive Attacks.
- Explain types of Attacks.
- Discuss Goals of Security.
- Explain Security Goals.
- Explain Euclidean Algorithm.
- Difference between Substitution and Transposition techniques.
- Explain Substitution and Transposition techniques.
- Explain Caesar Cipher.
- Difference between Monoalphabetic and Polyalphabetic substitution techniques.
- Explain Monoalphabetic and Polyalphabetic substitution techniques.
- Explain Vigenere Cipher.
- Any example on Playfair cipher and Hill cipher.
- Explain with examples Keyed and Keyless Transposition Ciphers.

## Module 2: Symmetric and Asymmetric Key Cryptography

- Explain Principles of Block Cipher.
- Explain Data Encryption Standard (DES) algorithm.
- Explain ECB and CBC modes of block cipher.
- Explain Cipher Feedback (CFB) and Output Feedback (OFB) modes of block cipher.

## Module 3: Cryptographic Hash Functions

- Explain Cryptographic Hash Functions.
- Explain Properties of Secure Hash Function.
- Explain MD5 and SHA-1 algorithms.
- Explain MAC, HMAC and CMAC.